
Validation Paper. Convection Heat Transfer from Tube Banks in Cross Flow

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1 Introduction

Heat exchangers play an essential role in nearly all technologies in the industrialized world. They are widely used in space heating, refrigeration, air conditioning, power stations, chemical plants, petrochemical plants, petroleum refineries, natural-gas processing, sewage treatment, and electronics. The tube bank is one of the most common in heat exchanger designs.

Tube banks are typically arranged in either in-line or staggered configurations and are described by dimensionless transverse, longitudinal, and diagonal pitch ratios. Khan et al. [1] developed an analytical model for the heat transfer from tube banks in terms of longitudinal and transverse pitches, Reynolds numbers, and Prandtl numbers.

2 Purpose

The purpose of this study is to validate **FLUENT**'s ability to predict the heat transfer characteristics of a tube bank in cross flow. This study also aims to validate the accuracy of the analytical model developed by Khan et al. with numerical results.

3 Problem Description

An 8 x 7 bank of tubes with diameter 16.4 mm is placed in cross flow of air with velocity $U_{app} = 6$ m/s and $T_a = 15^\circ\text{C}$ as shown in Figure 1. The tubes are in a compact, staggered arrangement and are held at constant temperature $T_s = 70^\circ\text{C}$. The goal is to analyze how flow speed, expressed through the Reynolds number, influences the Nusselt number across the tube bank.

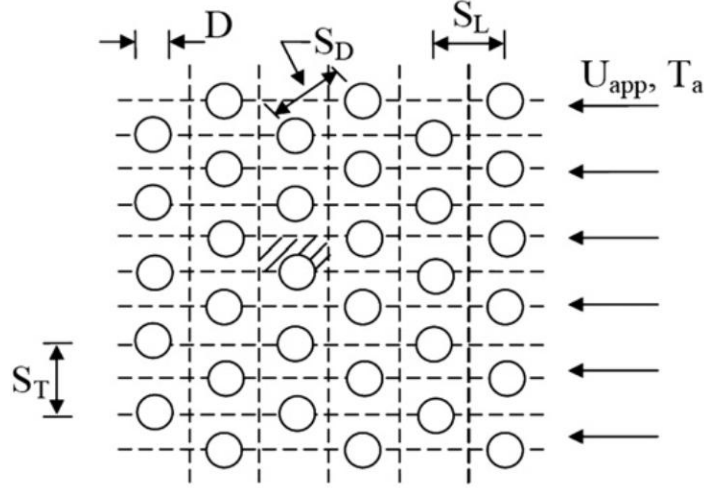


Figure 1. Schematic of staggered arrangement.

Table 1. Data used by Incropera and DeWitt [2] for staggered tube bank.

Quantity	Dimension
Tube diameter, D (mm)	16.4
Longitudinal pitch, S_L (mm)	20.5
Transverse pitch, S_T (mm)	20.5
Number of tubes (Staggered)	8×7
Approach velocity, U_{app} (m/s)	6
Thermal conductivity of air, k_{air} (W/m-K)	0.0254
Density of air, ρ (kg/m ³)	1.217
Specific heat of air, c_p (J/kg-K)	1007
Kinematic viscosity, ν (m ² /s)	14.82×10^{-6}
Prandtl number, Pr (Air)	0.701
Ambient temperature, T_a (°C)	15
Tube surface temperature, T_s (°C)	70

Khan et al. determined the coefficient C_1 in the following equation in terms of longitudinal and transverse pitch ratios for a staggered tube bank arrangement. For isothermal boundary conditions, the coefficient is given by:

$$C_1 = \frac{0.61 S_T^{0.091} S_L^{0.053}}{1 - 2 \exp(-1.09 S_L)} \quad (1)$$

Where S_T and S_L represent the transverse and longitudinal pitches, respectively. Eq. (1) is valid for $1.05 \leq S_L \leq 3$ and $1.05 \leq S_T \leq 3$. The dimensionless heat transfer coefficient in terms of C_1 , Re_D and Pr numbers can be written as:

$$Nu_D = \frac{hD}{k_{air}} = C_1 Re_D^{\frac{1}{2}} Pr^{\frac{1}{3}} \quad (2)$$

3.1 Fluid Properties

The properties of the fluid are assumed to be constant, as shown in Table 1.

3.2 Flow Physics

The Reynolds number is based on the tube diameter and the free stream velocity. Different values of Re_D (100, 500, 1000) in the simulation were obtained by changing the approach flow velocity.

These values were selected because Khan et al.'s data were extrapolated over a range of about $Re_D = 10^2$ to 10^5 . Moreover, these lower Reynolds numbers allow for use of a laminar model.

$$Re_D = \frac{UD}{\nu} \quad (3)$$

3.3 Boundary Conditions

A no-slip condition was applied to all tube walls, while the inlet and outlet were modeled as periodic with specified mass flow. A symmetry boundary condition was applied to the top and bottom boundaries. The boundary conditions used in the simulation are illustrated in Figure 2.

4 Grid

A 6006 cell, 12206 face, 6220 node mesh was used in this validation study as shown in Figure 4. To validate grid convergence using the algorithm in Celik et al. [3], three grids of increasing refinement were used.

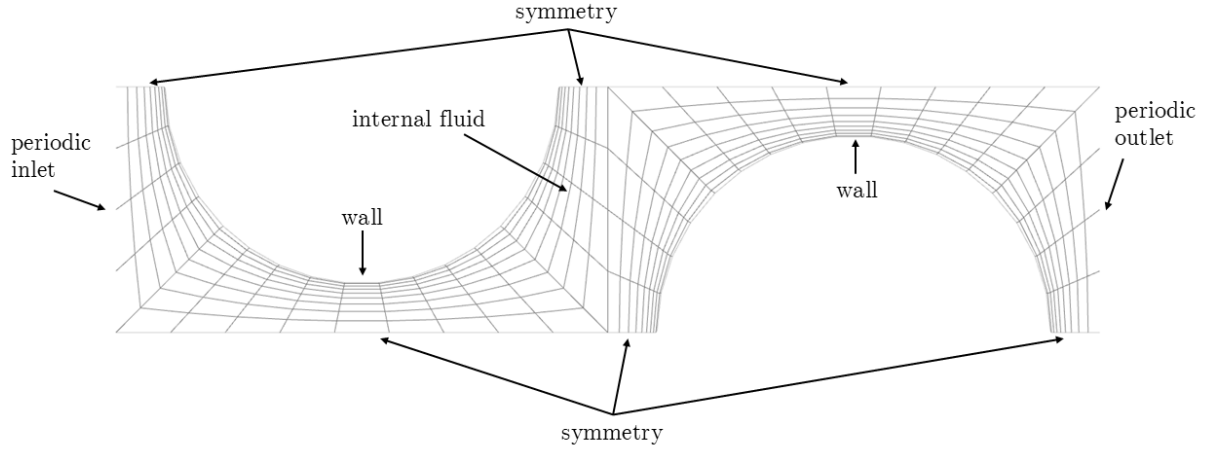


Figure 2. Coarse grid with boundary conditions. 305 cells, 656 faces, 355 nodes.

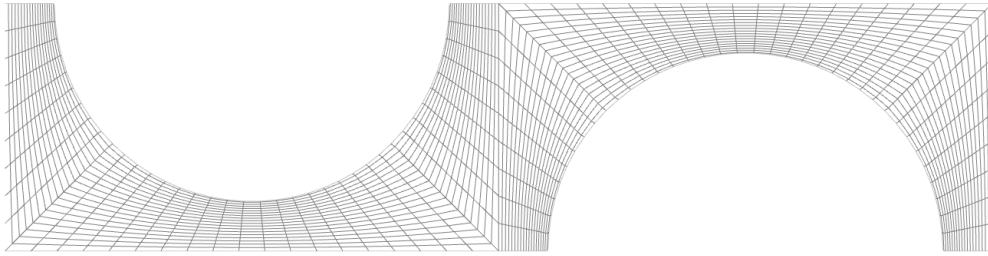


Figure 3. Finer grid. 1406 cells, 2915 faces, 1510 nodes.

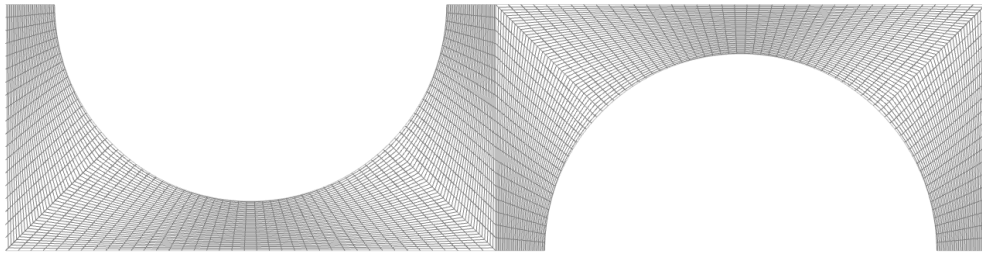


Figure 4. Finest grid, used in this study. 6006 cells, 12206 faces, 6220 nodes.

5 Case Setup

The **FLUENT** case was set up using constant fluid properties and the boundary conditions described in Section 3.3. Three runs were conducted using varying approach velocities to produce the Reynolds numbers shown in Table 2.

Table 2. Run Conditions.

Run	1	2	3
Reynolds Number, Re_D	100	500	1000

6 Calculation

For velocity-pressure coupling, the Coupled scheme was used. For spatial discretization, least squares cell based was used for the gradient. Second order discretization was used for pressure. Second order upwind was used for momentum and energy. Pressure and momentum relaxation factors were set to 0.5, while density and body force relaxation factors were set to 1.

Convergence criteria were set to 1×10^{-6} for continuity, x-velocity, and y-velocity. Finally, the solution initialization was performed using the hybrid method.

7 Results

For each Reynolds number, the total heat transfer rate across both half-tubes was multiplied by 8 x 7 to represent the heat transfer across the entire bank.

Using Newton's Law of Heating and Cooling, a representative overall heat transfer coefficient was solved for.

T_s is the tube surface temperature and T_a is the ambient air temperature. A is the unit area of each tube where $L = 1\text{m}$.

$$Q = hA(T_s - T_a), \quad A = \pi DL \quad (4)$$

Then the Nusselt number was solved for with respect to tube diameter, heat transfer coefficient, and the thermal conductivity of the air.

$$h = \frac{Q}{A(T_s - T_a)} \quad (5)$$

$$Nu_D = \frac{hD}{k_{air}} \quad (6)$$

Contour plots of temperature around two representative tubes are shown in Figures 5, 6, and 7.

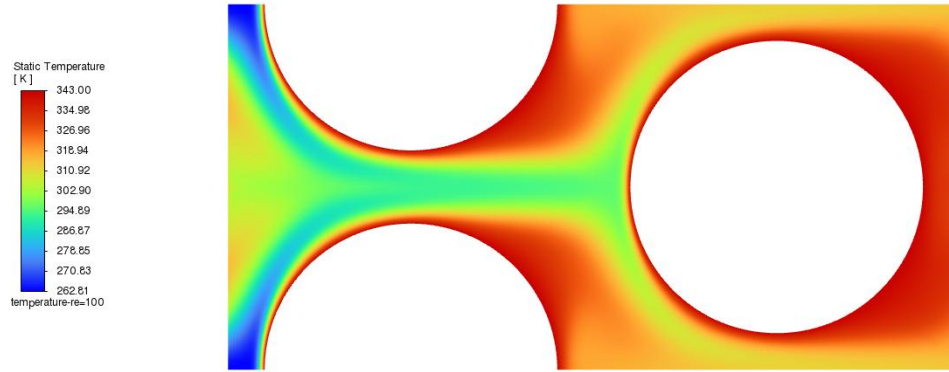


Figure 5. Static Temperature Contours ($Re_D = 100$).

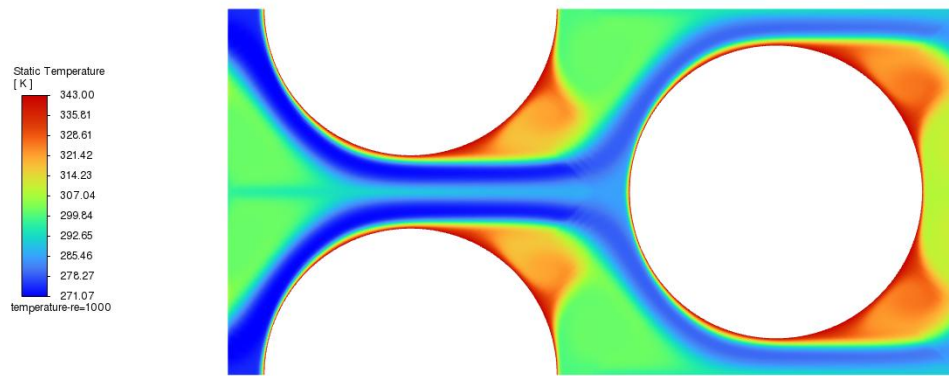


Figure 6. Static Temperature Contours ($Re_D = 500$).

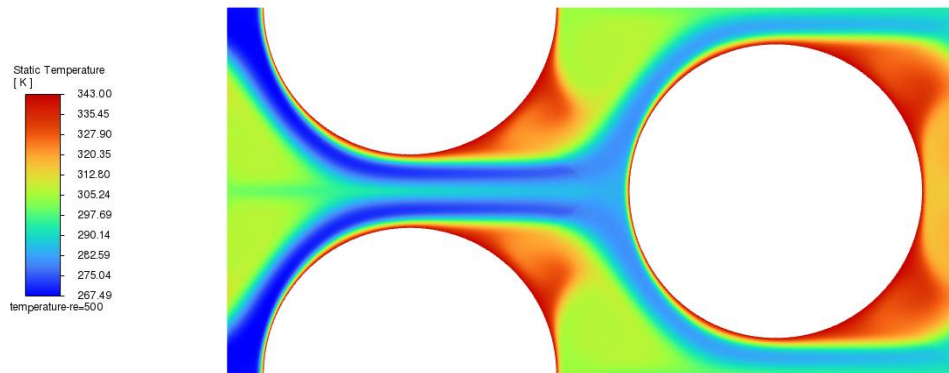


Figure 7. Static Temperature Contours ($Re_D = 1000$).

The results are compared with those of Khan et al.'s using Eq. (2) in Figure 8.

The heat transfer increases linearly with Reynolds number on the logarithmic scale. The log-log plot in Figure 8 of Nusselt number vs. Reynolds number shows a power-law relationship, consistent with theoretical expectations. The results from the three simulations are shown in Table 3.

The error increases as the Reynolds number rises. This is expected, as the flow approaches transitional or turbulent regimes, making the laminar model used in **FLUENT** inaccurate. For these reasons, data is not extrapolated far past $Re_D \approx 3000$ as a rule of thumb for laminar to turbulent transition in external flow.

The present analysis gives higher results (around 17%) than the analytical results from Khan et al. However, for strongly laminar flow regimes, the error is significantly lower.

Table 3. Nusselt number results for three simulations.

	$Re_D = 100$	$Re_D = 500$	$Re_D = 1000$
Nu_D	13.73	34.16	46.042

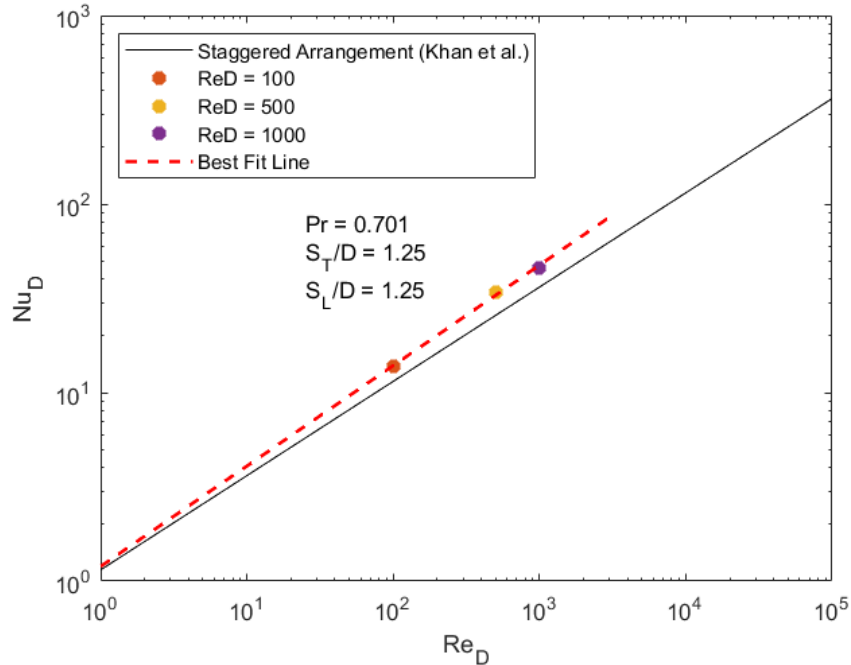


Figure 8. Comparison of present analysis to analytical approach from Khan et al.

8 Grid Independence

To verify grid independence, heat transfer rate as a function of grid resolution was recorded in three separate simulations, all with $Re_D = 500$. The data in Table 4 were used to perform the procedure for estimation of discretization error outlined in Celik et al. The representative grid sizes are based on the data in Section 4.

Table 5. Results from grid independence simulations.

Representative Grid Size, h	Heat Transfer Rate, Q (kW)
0.3605	8.35
0.2426	8.72
0.1673	8.78

The Richardson extrapolation algorithm yielded the following results.

Table 6. Results from Richardson extrapolation

Apparent order, p	5.083
Approximate relative error	0.64%
Extrapolated error	0.11%
Fine-grid GCI	0.14%

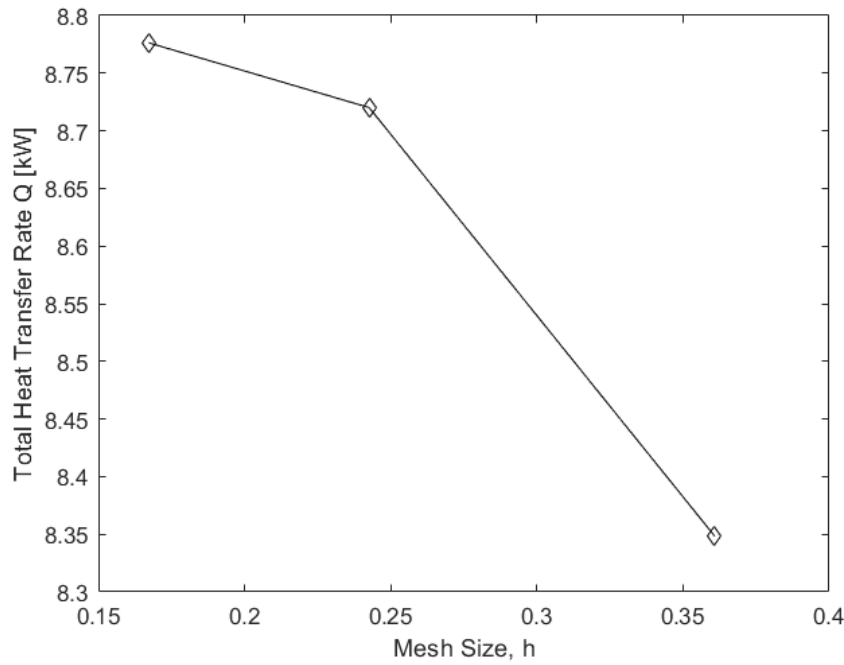


Figure 9. Heat Transfer Rate vs Grid Size at $Re_D = 500$.

9 Conclusion

FLUENT has been validated for the classical example of laminar external flow of air around a bank of tubes. Despite a moderate overestimation of the tube bank's heat transfer characteristics, the simulation results align closely with the analytical model devised by Khan et al. for strongly laminar flow.

10 References

- [1] W.A. Khan, J.R. Culham, M.M. Yovanovich, Convection Heat Transfer from Tube Banks in Crossflow: Analytical Approach, University of Waterloo, 2006.
- [2] F.P. Incropera, D.P. DeWitt, Introduction to Heat Transfer, John Wiley and Sons, Inc., New York, 2002.
- [3] I.B. Celik et al., Procedure for Estimation and Reporting of Uncertainty, West Virginia University, 1993